Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/24/2025
PI Name:	Ivkovic, Vladimir Ph.D.		
Project Title:	Characterizing the Baselines of Sleep Quality, Cognitive / Operational Performance, Immune Function, and Intracranial Fluids for Deep Space Expeditions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02129	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	04/22/2019	**End Date:**	04/30/2031
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	11	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	FY24 Report: Nina Vujovic, Ph.D. is no longer with the project. JoAnna Pollonais, Ph.D., Heather Quirarte, Ph.D., Brian Crucian, Ph.D., and James Ehlers, Ph.D. have been added as Co-Investigators. The Principal Investigator (PI) for this investigation changed from Dr. Quan Zhang to Dr. Vladimir Ivkovic in August 2022. Dr. Zhang remains on the project as a Co-Investigator. Dr. Matthias Basner and Dr. Alexander Stahn, from the University of Pennsylvania, were also added to the project as Co-Investigators (Ed., 9/2/24).		
COI Name (Institution):	Strangman, Gary E. (Massachusetts General Hospital / Harvard Medical School) Zhang, Quan (Massachusetts General Hospital / Harvard Medical School) Kimberly, William T. (Massachusetts General Hospital / Harvard medical School) Pollonais, JoAnna M.S. (Massachusetts General Hospital / Harvard Medical School) Spielmann, Guillaume (Louisiana State University / School of Kinesiology) Quiriarte, Heather M.Sc. (Louisiana State University / School of Kinesiology) Basner, Mathias M.D., Ph.D. (University of Pennsylvania) Stahn, Alexander (University of Pennsylvania) Crucian, Brian E. Ph.D. (NASA Johnson Space Center)		

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Task Description:	Behavior Associated with Sleep & Immunity in Space (BASIS) [formerly: Characterizing the Baselines of Sleep Quality, Cognitive / Operational Performance, Immune Function, and Intracranial Fluids for Deep Space Expeditions] Sleep is a central physiological regulator of cognitive / behavioral, neurophysiological, and immune functions. The study of sleep quality and duration on orbit may thus yield important insights into etiology and mechanisms of adverse cognitive/behavioral and immunological responses during long duration deep space exploration missions. Deep space and planetary expeditions will expose crewmembers to many risks that could adversely affect mission outcomes. We will use analog-tested cognitive (Cognition) and operational (ROBoT-r) tests for performance assessment and data/sample share for a range of sleep quality and quantity measures, as well as characterizing neuroendocrine and immune functions. By looking across ISS missions of different lengths, we aim to characterize baselines of sleep, cognitive, operational, immune, function. We will collect data on crewmembers participating in Complement of Integrated Protocols for Human Exploration Research (CIPHER), including flyers on short-duration (<3 months), nominal-duration (~6 months) and long-duration (~12 months) missions aboard the International Space Station (ISS). We also plan to study demographically matched control subjects in Earth-based analogs (e.g., the Human Exploration Research Analog, or HERA). Within Complement of Integrated Protocols for Human Exploration Research (CIPHER) project, BASIS is a part of an international project comprising of three synergistic studies lead by Drs. Basner, Stahn, and Ivkovic, respectively. Our specific aims are: (SA1) Characterize cognitive and operational task performance changes as a function of ISS mission duration; (SA2) Characterize cognitive and operational task performance changes as a function of sleep status; (SA3) Characterize cognitive and operational task performance changes as a function of immune activation; (SA4) Characterize cognitive and operational task performance changes as a function of stress; and (SA5) Investigate the interactions between behavior, stress, immunity and sleep. The outcomes of the study will contribute to quantification of crew health and performance risks associated with human spaceflight, and aid in development of technologies for monitoring and mitigating crew health and performance. BASIS-PNI (Psychoneuroimmunology) Supplement: A supplement to the parent BASIS project, titled Behavior Associated with Sleep & Immunity in Space – Psycho-Neuro-Immunology (BASIS-PNI) was approved by the Human Factors and Behavioral Performance (HFBP) element in June 2020 (around the height of the first COVID wave). The supplement seeks to accelerate the identification and characterization of psychoneuroimmunological interactions that contribute to risks of adverse cognitive/behavioral conditions or mental disorders during exploration class missions. During such missions, crew members will be exposed to numerous stressors including microgravity, prolonged periods of isolation and confinement, environmental extremes, operational demands, and psychosocial challenges. One of the acutely visible and persistent signs of microgravity-induced cephalad fluid shift is facial edema and deformation that consequently reduces an individual's facial expressiveness and capacity to convey non-verbal affective cues. Since facial expression of affective content presents a fundamental aspect of social interaction and maintenance of interpersonal relations, any changes or limitations in this capacity during spaceflight could exacerbate the negative communication or psychological effects of isolation and confinement, and hence contribute to the risk of developing behavioral or psychiatric disorders (BMed). In addition, sustained exposure to psychological stressors, confinement, and isolation are all known to reduce immune competency and promote systemic inflammation and alter facial expressiveness on Earth and in spaceflight. The state of chronic low-grade inflammation is in fact associated with decrements in both physical and psychological functions. Elevation in pro-inflammatory cytokines including IL-1beta, IL-6 and TNF-alpha are associated with increased anxiety and altered mood in healthy adults, as well as reduced processing speed and overall cognition in older adults. It remains unknown whether spaceflight-relevant isolation, confinement, and exposure to operational stressors are associated with inflammation-mediated changes in neurocognitive function and affective processing in the astronaut-like population, nor how these neurocognitive deficits may affect the ability of crewmembers to interpret each-others' altered facial expressiveness due to microgravity-induced facial edema. BASIS-PNI-BR (Bed Rest) Supplement: A supplement to the parent BASIS project, titled Behavior Associated with Sleep & Immunity in Space – PsychoNeuroImmunology in Head Down Tilt Bedrest (BASIS-PNI-BR) was approved by the Human Factors and Behavioral Performance (HFBP) element in April 2024. The primary goal of this supplement is to characterize the relationship between perceived emotional valence based on previously recorded facial expression photographs—collected head up/head down tilt— and immune function before, during, and after exposure to 60-day head down tilt (HDT) bedrest (BR)—conducted at DLR :envihab facility. This data from HDTBR analog would be synergistic to the data collected in the C7 HERA campaign which is a part of the PNI supplement.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	BASIS: Successful completion of this project will be a milestone in spaceflight behavioral, neurophysiological, and immune investigation. It will be the largest study of operational performance in space, as well as the first to conduct in-flight resting-state and task-related functional brain imaging. In addition, it will be the first study to probe the relationships and interactions between behavioral, neurophysiological, sleep, and immune functioning. The use of three different length missions will help us to identify effects or trends that may extrapolate towards longer missions, with the known caveats of such extrapolation. The findings will be important on Earth as well, by helping to better understand the complex inter-relationships between sleep, brain physiology, immune function, and cognitive performance. BASIS-PNI: Findings from the PNI study will directly contribute to identification of key threats to behavioral health (BMed3), development of measures for monitoring behavioral health (BMed2), and understanding interactions between immune, psychological and perceptual factors in behavioral health associated with long duration missions. BASIS-PNI-BR: Findings from this study will further characterize the effects of HDT bedrest on neurobehavioral and immunological function, and thus directly contribute to identification of key threats to behavioral health, development of measures for monitoring behavioral health, and understanding interactions between immune, psychological and perceptual factors in behavioral health associated with long duration missions. By using the SAFER survey – developed as a part of the BASIS-PNI project and tested in HERA, this study will further validate this measure in another

spaceflight analog – HDT BR – thus expanding available tools for assessment of health and human performance in spaceflight.

Task Progress:

We continued to successfully collaborate within the integrated CIPHER-BASIS project consisting of a 3-partite Spatial Cognition project co-led by Drs. Basner, Stahn, and Ivkovic as Co-PIs. We have continued to collect data during research operations pre-, in-, and post-flight within the Spatial Cognition project. As of April 2025, we have collected preflight data on n=7 participants and inflight data on n=4 participants. Within the BASIS-PNI supplement, we completed the data acquisition and analysis pipeline for HUT/HDT 2D facial images and 3D facial models and finalized the Spaceflight Analog Facial Emotion Recognition (SAFER) computer-based survey. We are continuing to analyze images and models on N=21 participants and have completed data collection within NASA Human Exploration Research Analog (HERA) C7 (M1-M4; n=16 participants). This data is currently being analyzed. Finally, we successfully completed BASIS-PNI-BR data collection within the 1st mission of the DLR / :envihab 60-day head-down tilt bedrest (HDT BR) campaign. This data is currently being analyzed as we prepare for data collection in the upcoming mission 2 of the campaign. Our work on all three studies continues as planned.
Progress & Achievements in the Past Year: During the 6th year of this project, we focused on the following major tasks: • Completion of planned activities carrying over from Year 5. • Continuation of integration, recruitment, and data collection on CIPHER-BASIS. • Completion of validation and implementation of data acquisition pipeline for BASIS-PNI supplement. • Completion of Massachusetts General Hospital data collection and development of SAFER survey within BASIS-PNI supplement. • Completion of Data collection within HERA C7 within BASIS-PNI. • Integration and data collection of BASIS-PNI-BR supplement project. • Educational activities for students on CIPHER-BASIS, BASIS-PNI, and BASIS-PNI-BR project/supplements. • Presented 1 talk (CIPHER-BASIS-PNI) and 2 posters (CIPHER-BASIS, CIPHER-BASIS-PNI-BR) at the 2025 NASA Human Research Program Investigators' Workshop (HRP IWS).

Completion of planned activities carrying over from Year 5 We first completed the activities that carried over from year 5: • Continued recruiting (via ICBs) and collecting baseline as well as in-flight data collection for CIPHER crewmember participants.

Continuation of integration, recruitment, and data collection on CIPHER-BASIS During the report period, we continued working with ROI and CIPHER collaborators to recruit additional crewmember participants for the CIPHER-BASIS project, as well as collect data in pre-flight, and in-flight phases of International Space Station (ISS) missions. • Updated data sharing plan (DSP) and data sharing agreement (DSA), preparing for round 4 of Institutional Review Board (IRB) updates. • Third round of IRB integration with Basner/Stahn team completed. • Amendments to IRB on updated procedures for ROBoT completed. • Integration with other CIPHER projects continuously being worked out via biweekly calls with Bruce Keller, Scott Humbert, et al./ROI. • Contributed to and completed SRD editing and TRR preparation. • Consented first 5 subjects as of April 2025. • Collected preflight ROBoT data on 7 subjects, inflight data on 3 subjects, and postflight data on 3 subjects as of April 2024. • Consenting, preflight, inflight data collection continuing.

Completion of validation and implementation of data acquisition and analysis pipeline for BASIS-PNI supplement During the past year, we completed validation and implemented the data acquisition and analysis pipeline for 2D facial images and 3D facial models in various degrees of head-up tilt (HUT) and head-down tilt (HDT). These activities included test/re-test reliability of 2D image and 3D model acquisition, and processing consisting of 3D facial scanning plus inter-image registration, quantification, and visualization. In addition, components of the HUT/HDT testing apparatus for head/neck adjustment and HD camera positioning during HUT/HDT, tilt table, and Shining 3D Einscan-H 3D scanner procedures were tested and integrated in the data collection protocol.

Completed Massachusetts General Hospital data collection and development of SAFER survey within the BASIS-PNI supplement project During the past year, we collected 2D images and 3D facial models from N=21 participants. For each participant, 2D images and 3D models were acquired in +50o HUT, 0o (supine), -6o HDT, and -12o HDT orientations. A total of 168 3D images were collected. Each orientation was maintained for 10 minutes. Images/models were collected at the beginning (~1st minute), in the middle of (~5th minute), and at the end (9th minute) of orientation for monitoring temporal change in facial features for subsequent analyses. At each orientation and time point, 2D images were acquired as participants displayed facial expressions conveying 5 emotions – disgust, anger, neutral, sadness, and happiness – in a randomized order. We also completed development of the SAFER survey which included integration of 2D images of facial expressions collected at the end of each HUT/HDT orientation into an image database. A Python-based SAFER code was optimized to display each image for 1000ms in a randomized order, followed by a Likert scale which the survey taker is prompted to use to assign facial emotional expression to the 2D facial image was displayed. SAFER takes approximately 10 minutes to complete.

Completed data collection within HERA C7 within BASIS-PNI In the past year, we completed data collection for the BASIS-PNI supplement study in HERA C7. We also maintained collaboration with Dr. Brian Crucian at NASA that enabled the collection and data sharing of additional immunology data during HERA missions. Starting in January 2024, BASIS-PNI data collection was started in HERA C7 M1 and continued through HERA C7 M4, thus marking the completion of data collection for C7 (M1-M4). Data was collected on a sample of N=16 HERA crew, 50% female, 37±5.5 years of age. As a part of the BASIS-PNI in HERA C7, we are collecting the following data/samples: • Biological: Resting saliva sample to measure stress responses--cortisol, DHEA-S, epinephrine, norepinephrine, dopamine, alpha amylase, lysozyme, and chromogranin A (ELISA); Resting serum sample to measure immune and stress response--circulating cytokines (22-plex). • Surveys: Emotional state (Positive Affect Negative Affect Scale; PANAS), anxiety (State Trait Anxiety Index/STAI), and Pittsburgh Sleep Quality Index (PSQI). All data are being collected before (MD-1), during (MD23, MD37), and after (MD+1) missions within HERA C7. The data analyses are currently underway.

Integration and data collection of BASIS-PNI-BR supplement project During the past year, the BASIS-PNI-BR supplement has been integrated into the 60-day BR sensorimotor study at DLR/:envihab. As a part of the BASIS-PNI-BR in DLR/:envihab, we are collecting the following data/samples: • Biological: Stress and immune response measures will be used in conjunction with psychometric data; Plasma samples from all the subjects will also be analyzed in duplicate for cytokines and biomarkers of inflammation and stress including cortisol, DHEA-S, epinephrine, norepinephrine, dopamine, alpha amylase, lysozyme, and chromogranin (NULISA-seq); All saliva samples will be collected using the non-invasive commercially-available Salimetrics polyester Oral Swab (SOS) technology (Salimetrics, CA, USA) which has been validated for the measurement of salivary hormones, sIgA and alpha-amylase. •

	Surveys: Emotional state (Positive Affect Negative Affect Scale; PANAS), anxiety (State Trait Anxiety Index/STAI), General Health Questionnaire (GHQ), Pittsburgh Sleep Quality Index (PSQI), Spaceflight Analog Facial Emotion Recognition (SAFER). Based on the approved SOW for this supplement, during the report period we achieved the following milestones: • Worked with DLR and NASA ROI to implement the study in the first mission of the :envihab 60-day HDT bedrest study. • Implemented data sharing plans with DLR. • Ongoing collaboration with Dr. Guillaume Spielmann at Louisiana State University (LSU) at NASA to enable analyses of immunology data and IRB integration. • Completed data collection during the first mission of the campaign from September through December 2024: Before (BDC-12 to BDC-1 time window), during (HDT12/HDT45 to HDT50/HDT60 time window), and after (R+0 and R+15 window) BR mission. • Data was collected on n=12 participants (8F, 4M) aged 32±5 years. • Conducted quality checks on incoming data during the campaign (all data nominal). • Preliminary data analyses, and coordinating with SMC ROI and DLR teams are ongoing. • Coordinating for the upcoming 2nd mission of the campaign scheduled to start in May 2025. Educational activities for students CIPHER-BASIS Project • We continued to train 2 graduate student interns on the CIPHER project; 1 undergraduate student who was previously working on the CIPHER project completed their degrees during the report period. • Students participated in the development of research procedures and protocols. • The students were included as co-authors and participated in the development and delivery of the CIPHER-BASIS poster presentation at the 2025 HRP IWS. BASIS-PNI Project • We continued to work with 4 student interns (1 graduate, and 3 undergraduates) on the BASIS-PNI study supplement project. • The students participated in the revision of IRB protocols, development of research procedures and protocols, design development for laboratory research apparatus, quality control and pilot testing, protocol optimization, data collection and analyses, as well as design/development of the SAFER survey. • The students participated in the literature review and preparation of a review manuscript focusing on psychoneuroimmunological aspects of neurobehavioral changes in spaceflight; the manuscript is in preparation. • The students were included as co-authors and participated in the development and delivery of the BASIS-PNI poster presentation at the 2025 HRP IWS. BASIS-PNI-BR Project: • One graduate student contributed to the preparation of the BASIS-PNI-BR supplement project. • The student participated in the preparation of study protocols, hardware/software preparation for the study, IRB document preparation, and study integration steps.
Bibliography Type:	Description: (Last Updated: 05/13/2026)
Abstracts for Journals and Proceedings	Ivkovic V, Basner M, Stahn AC, Spielmann G, Pollonais J, Zhang Q, Strangman GE. "Temporal nature of cognitive and visuospatial brain domain changes during long-duration low-Earth orbit missions: Baselines of sleep, cognitive/operational performance, and immune function (Project C)." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Ivkovic V, Ehlers J, Pollonais J, Spielmann G, Quiriarte H, White BM, Zhang Q, Strangman GE. "BASIS-PNI: Behavior associated with sleep & immunity in space – Psychoneuroimmunology." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Ivkovic V, Pollonais J, Spielmann G, Quiriarte H, Piechowski S, Moestl S, Mulder E, Strangman GE. "BASIS-PNI-HDBR: Behavior associated with sleep & immunity in space – Psychoneuroimmunology in head down tilt bedrest – Preliminary update." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025